1. $\dfrac{2x-5}{x-2} < 1$

$\Rightarrow \dfrac{2x-5}{x-2} - 1 < 0$

$\Rightarrow \dfrac{2x-5-(x-2)}{x-2} < 0$

$\Rightarrow \dfrac{x-3}{x-2} < 0$

→ Pembilang: $x-3=0$, $x=3$ — Penyebut: $x-2=0$, $x=2$

→ $+ \quad - \quad +$ (2, 3) — uji coba $x=0$: $\dfrac{(-)}{(-)} = (+)$

Hp $= 2 \leq x < 3$

$(2,3)$

MUTLAK

$|a| = \begin{cases} a, & a \geq 0 \\ -a, & a < 0 \end{cases}$

1. $|7x-4| = 2$ $\quad \begin{cases} 7x-4, & 7x-4 \geq 0 \\ -(7x-4), & 7x-4 < 0 \end{cases}$

I. $7x-4 = 2$ → $7x-4 \geq 0$

$7x = 6$ → $7\left(\frac{6}{7}\right) - 4 = 2 \geq 0$

$x = \frac{6}{7}$

II. $-(7x-4) = 2$ → $7x-4 < 0$

$7x = 2$ → $7\left(\frac{2}{7}\right) - 4 = -2 < 0$

$x = \frac{2}{7}$

Hp $= \left\{ \frac{6}{7}, \frac{2}{7} \right\}$

☆ CARA LAIN

$|7x-4| = 2$

$-2 \quad 2$

I. $7x-4 = 2$, $7x = 6$, $x = 6/7$

II. $7x-4 = -2$, $7x = 2$, $x = 2/7$

2. $|3x-2| = |5x+4|$

$\sqrt{a^2} = |a| \rightsquigarrow a^2 = |a|^2$

→ $(3x-2)^2 = (5x+4)^2$

$9x^2 - 12x + 4 = 25x^2 + 40x + 16$

$0 = 16x^2 + 52x + 12$

$0 = 4x^2 + 13x + 3$

$\frac{1}{4} \cdot (4x+12)(4x+1)$

$(x+3)(4x+1)$

$x = -3 \;\vee\; x = -\frac{1}{4}$

3. $|3x+1| < 4$

$\left|3\left(x+\frac{1}{3}\right)\right| < 4$

$|3| \cdot \left|x+\frac{1}{3}\right| < 4$

$3 \cdot \left|x+\frac{1}{3}\right| < 4$

$\left|x+\frac{1}{3}\right| < \frac{4}{3}$

$\left|x-\left(-\frac{1}{3}\right)\right| < \frac{4}{3}$

$4/3 \quad 4/3$

$-\frac{1}{3} - \frac{4}{3} = -\frac{5}{3} \qquad -\frac{1}{3} \qquad -\frac{1}{3} + \frac{4}{3} = 1$

$\therefore -\frac{5}{3} < x < 1$ atau $\left(-\frac{5}{3}, 1\right)$

☆ CARA LAIN

$|3x+1| < 4$

$|\ldots| < 4$ maka $-4 \quad 4$

$-4 < 3x+1 < 4$

$-5 < 3x < 3$

$-\frac{5}{3} < x < 1$

4. $|x+4| \geq 2$

$|\ldots| \geq 2$ maka $-2 \quad 2$

$x+4 \leq -2$ atau $x+4 \geq 2$

$x \leq -6 \qquad x \geq -2$

$\therefore (-\infty, -6] \cup [-2, +\infty)$

☆ $|x| < a \longrightarrow -a < x < a$

$|x| > a \longrightarrow x > a \cup x < -a$

5. $\dfrac{1}{|2x-3|} > 5$

I. Penyebut $\neq 0$

$2x-3 \neq 0$

$2x \neq 3$

$x \neq \frac{3}{2}$

II. krn Mutlak ≥ 0 v penyebut $\neq 0$

maka $|2x-3| > 0$, jadi bs kali silang

$1 > 5 \cdot |2x-3|$

$\frac{1}{5} > |2x-3|$

$|2x-3| < \frac{1}{5}$

$-\frac{1}{5} < 2x-3 < \frac{1}{5}$

$< 2x <$

PERTIDAKSAMAAN SEGITIGA

$||a+b| \leq |a|+|b|$

*≡

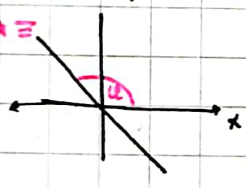
m > 0 m < 0 m = 0

ex :

$m_2 = 20$

$m_1 = 10$

*≡



$m = \tan \alpha$

α : sudut antara sb.x & garis, berlawanan arah jarum jam.

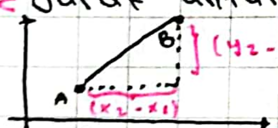
*≡ $l_1 \parallel l_2 \rightarrow m_1 = m_2$

*≡ $l_1 \perp l_2 \rightarrow m_1 \cdot m_2 = -1$

*≡ $m = \dfrac{y_2-y_1}{x_2-x_1}$

*≡ $y - y_1 = m(x - x_1)$

*≡ Jarak antara 2 titik



$J_{AB} = \sqrt{(y_2-y_1)^2+(x_2-x_1)^2}$

(phytagoras)

*≡ Titik tengah

$\left(\dfrac{x_1+x_2}{2}, \dfrac{y_1+y_2}{2}\right)$

pers. lingkr

$x^2+y^2=r$

$(x_1-a)(x-a)+(y_1-b)(y-b)=r^2$

* a & b = pusat lingk

*≡ Jarak Titik ke Garis

(tegak lurus)

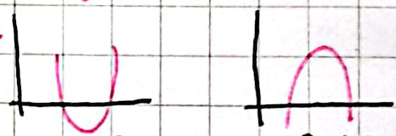
Jarak (d) antara titik (x_0, y_0) ke garis $ax+by+c=0$:

$d = \dfrac{|ax_0+by_0+c|}{\sqrt{a^2+b^2}}$

*≡ Persamaan Lingkaran

$r^2 = (x-x_0)^2 + (y-y_0)^2$

*≡



a > 0 a < 0

$x_p = \dfrac{-b}{2a}$

FUNGSI

fungsi

1. $h(x) = \dfrac{x^2-4}{x+2}$

Df $x+2 \neq 0$

$x \neq -2$

Rf $= \dfrac{\cancel{(x+2)}(x-2)}{\cancel{(x+2)}} = x-2,\ x \neq -2$

$y = x-2,\ x \neq -2$

$x < -2$ $x > -2$

$x-2 < -4$ $x-2 > -4$

$y < -4$ $y > -4$

$y \neq -4$

Range = $\{y,\ y \neq -4\}$

2. $h(x) = \sqrt{\dfrac{x(x-2)}{x-1}}$

I. Penyebut $\neq 0 \rightarrow x-1 \neq 0$

$x \neq 1 \rightarrow (-\infty,1) \cup (1,+\infty)$

II. Dalam akar ≥ 0

$\dfrac{x(x-2)}{x-1} \geq 0 \rightarrow$ Rf [y,x] $[0,1) \cup [2,+\infty)$

-- ++ -- ++

0 1 2

D(h) = $[0,1) \cup [2,+\infty)$

3. $f(x) = x^{\frac{1}{2}} + x^{-\frac{1}{2}} = \sqrt{x} + \frac{1}{\sqrt{x}} \cdot D_f = \{x > 0\}$

$g(x) = x^{\frac{1}{2}} - x^{-\frac{1}{2}} = \sqrt{x} - \frac{1}{\sqrt{x}} \cdot D_g = \{x > 0\}$

$D_{f/g} = ?$

$\left.\begin{matrix} x \geq 0 \\ x \neq 0 \end{matrix}\right\} x > 0$

↳ $g(x) = 0$

$\sqrt{x} - \frac{1}{\sqrt{x}} = 0$

$\sqrt{x} = \frac{1}{\sqrt{x}}$

$x = 1$

maka

(lg di pakel diambil yg)

$D_{f/g} = (D_f \cap D_g) \setminus \{x \mid g(x) = 0\}$

$= \{x > 0\} \setminus \{x = 1\}$

$= (0,1) \cup (1, +\infty)$ INI $D_{f/g}$ YG BENAR

lalu

$f/g\,(x) = \dfrac{f(x)}{g(x)}$

$= \dfrac{\sqrt{x} + \frac{1}{\sqrt{x}}}{\sqrt{x} - \frac{1}{\sqrt{x}}}$

$= \dfrac{\frac{x+1}{\sqrt{x}}}{\frac{x-1}{\sqrt{x}}} = \dfrac{x+1}{x-1}$ Benar

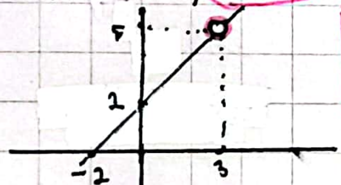
tapi

! $D_{f/g} = \{x \neq 1\}$ salah

2.3. ↳ $f(x) = \dfrac{x^2 - x - 6}{x - 3}$ → $x - 3 \neq 0$, $x \neq 3$

$= \dfrac{(x-3)(x+2)}{(x-3)}$

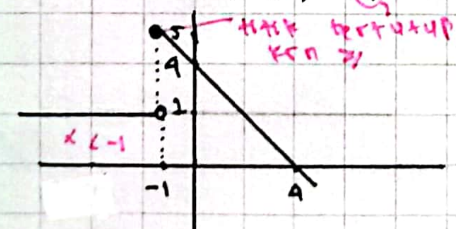
$y = x + 2$, $x \neq 3$



x	0	-2
y	2	0

↳ $g(x) = \begin{cases} 2 & x < -1, \\ 4 - x, & x \geq -1. \end{cases}$

tidak tertutup krn $\geq$

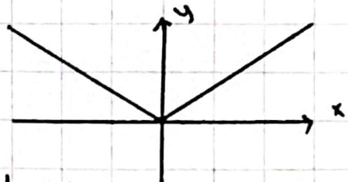


$x \geq -1$ →

x	0	4
y	4	0

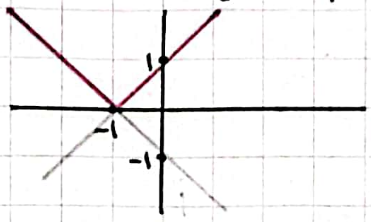
$x = -1 \rightarrow y = 5$

↳ $y = |x|$



$y = |x+1| = \begin{cases} x+1, & x+1 \geq 0 \\ -(x+1), & x+1 < 0 \end{cases}$

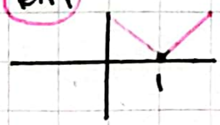
$= \begin{cases} x+1, & x \geq -1 \\ -x-1, & x < -1 \end{cases}$



I.	x	0	-1	$x \geq -1$
	y	1	0	
II.	x	0	-1	$x < -1$
	y	-1	0	

! jika dilihat grafik $y = |x|$ memiliki kemiripan dgn $y = |x+1|$ hanya di geser 1 satuan ke kiri

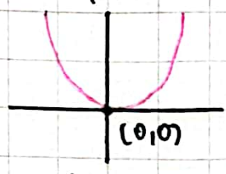
• $y = |x - 1|$

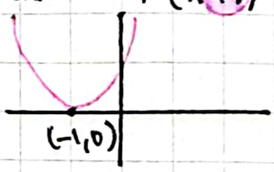


di geser 1 satuan ke kanan

• berlaku juga

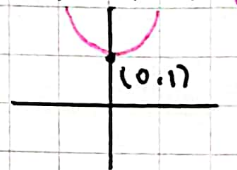
$y = x^2$ maka $y = (x+1)^2$





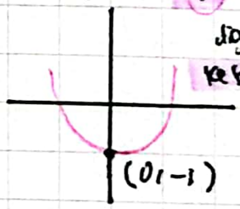
digeser 1 satuan ke kiri

• lalu $y = x^2 + 1$



digeser ke atas 1 satuan

$y = x^2 - 1$



digeser ke bawah

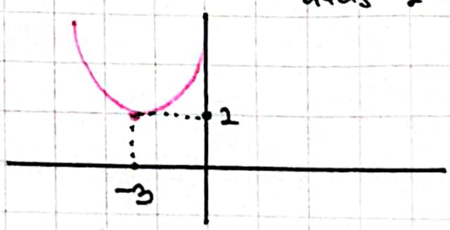
• pengaplikasian

$y = x^2 + 6x + 11$

$= (x^2 + 6x + 9) + 2$

$= (x+3)^2 + 2$

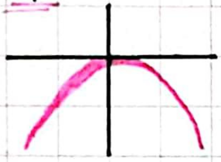
maka di geser kiri 3 satuan k atas 2 satuan.



PENCERMINAN

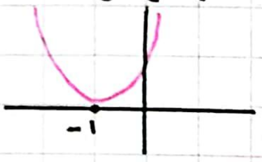
- thdp sb. x $\Rightarrow$ x tetap
  $y \to -y$

ex: ① $y = x^2$ ② $-y = x^2 \to y = -x^2$



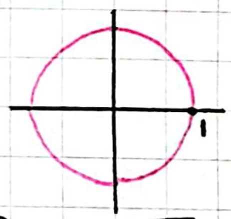
- thdp sb. y $\to$ y tetap
  $x \to -x$

ex: ① $y = (x+1)^2$ ② $y = (1-x)^2 = (x-1)^2$



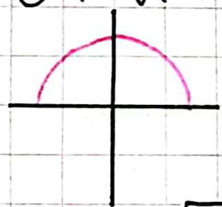
LINGKARAN

① $x^2 + y^2 = 1$



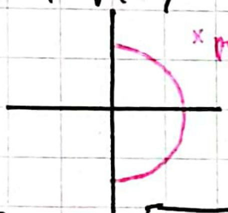
② $y = \sqrt{1-x^2}$

y positif



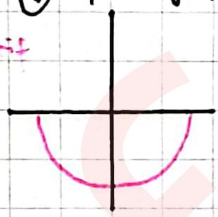
③ $x = \sqrt{1-y^2}$

x positif



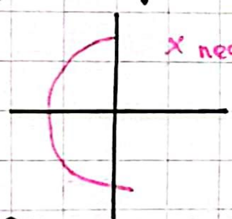
④ $y = -\sqrt{1-x^2}$

y negatif

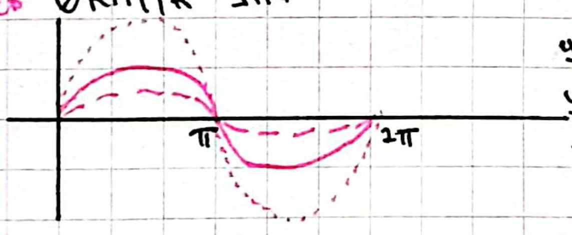


⑤ $x = -\sqrt{1-y^2}$

x negatif



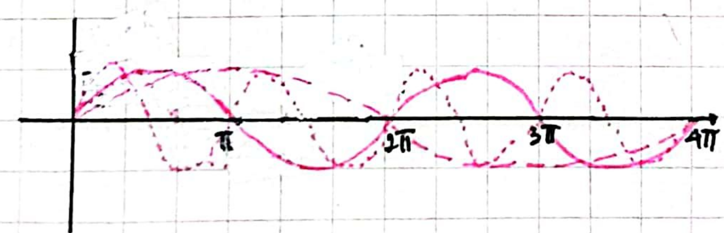
GRAFIK SIN



$y = 2 \sin x$

$y = \sin x$

$y = \frac{1}{2} \sin x$



$y = \sin x$ ——

$y = \sin 2x$ ····

$y = \sin \frac{1}{2}x$ – – –

invers

★ Ada 2 fungsi, apakah saling invers

ex: $f(x) = x^3$ & $f^{-1}(x) = x^{1/3}$ invers?

$(f \circ f^{-1})(x) = f(f^{-1}(x)) = f(x^{\frac{1}{3}}) = (x^3)^{\frac{1}{3}} = x$

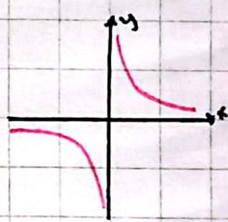
$(f^{-1} \circ f)(x) = f^{-1}(x^3) = (x^3)^{\frac{1}{3}} = x$ } saling invers

★ Ada 1 fungsi, apakah memiliki invers?

- fungsi satu-satu punya invers
  jika $f(x) = f(y)$ maka $x = y$
- jika $x \neq y$ maka $f(x) \neq f(y)$
  uji garis mendatar;
  grafik garis dipotong maks 1x
  oleh sebarang garis horizontal.

limit

1. $\lim_{x \to a} k = k$
2. $\lim_{x \to a} x = a$
3. $\lim_{x \to 0^-} \frac{1}{x} = -\infty$



4. $\lim_{x \to 0^+} \frac{1}{x} = +\infty$

   → fungsi polinomial

5. $\lim_{x \to a} p(x) = p(a)$ substitusiin doang
6. $\lim_{x \to -\infty \text{ or } +} p(x)$: lihat pangkat tertinggi

   ex: $\lim_{x \to -\infty} 7x^5 + 5x^4 - 2x^2 + 19$

   $= \lim_{x \to -\infty} 7x^5 = -\infty$

invers :

① $y = ax + b \to y' = \frac{x-b}{a}$

② $y = \frac{ax+b}{cx+d} \to y' = \frac{-dx+b}{cx-a}$

③ $f(\triangle) = \square$
$f^{-1}(\square) = \triangle$

→ fungsi rasional

7. $\lim_{x \to a} f(x) =$

I. Subtitusikan. (peny $\neq 0$)

II. peny $= 0$ — pemb $= 0$ → difaktorkan; pemb $\neq 0$ → grs bilangan

Ex: $\lim_{x \to 2} \frac{x-1}{x-2}$ → $+\infty$, $-\infty$, tdk punya limit

→ peny

$\lim_{x \to 2} x-2 = 2-2 = 0$

→ pemb

$\lim_{x \to 2} x-1 = 2-1 = 1 \neq 0$

garis bilangan (pembuat 0)

1. $x-1=0 \to x=1$
2. $x-2=0 \to x=2$

$+ + \quad - - - \quad + +$; 1, 2

$\lim_{x \to 2^+} \frac{x-1}{x-2} = +\infty$

$\lim_{x \to 2^-} \frac{x-1}{x-2} = -\infty$

∴ tidak punya limit

8. $\lim_{x \to +\infty} f(x)$ atau $\lim_{x \to -\infty} f(x) =$

pilih pangkat tertinggi pemb & peny

ex: $\lim_{x \to +\infty} \frac{4+2x-5x^3}{6x^2-10x^3}$

$= \lim_{x \to +\infty} \frac{-5x^3}{-10x^3} = \frac{1}{2}$

(!!) $\lim_{x \to +\infty} \frac{ax^m}{bx^n} =$ — $m < n \to 0$; $m > n \to +\infty$; $m = n \to \frac{a}{b}$

9. → limit memuat akar → dikalikan sekawan

ex: $\lim_{x \to 4} \frac{\sqrt{2x+1}-3}{x^2-16}$

lim pemb $\sqrt{2(4)+1} - 3 = 0$

lim peny $4^2 - 16 = 0$

→ $\lim_{x \to 4} \frac{\sqrt{2x+1}-3}{x^2-16} \times \frac{\sqrt{2x+1}+3}{\sqrt{2x+1}+3}$

$= \lim_{x \to 4} \frac{(2x+1)-9}{(x^2-16)(\sqrt{2x+1}+3)}$

$= \lim_{x \to 4} \frac{\cancel{2x-8}\ 2}{(x+4)\cancel{(x-4)}(\sqrt{2x+1}+3)}$

$= \lim_{x \to 4} \frac{2}{8(\sqrt{9}+3)} = \frac{1}{24}$

→ limit menuju tak hingga yg memuat akar

ex: ① $\lim_{x \to +\infty} \frac{\sqrt{x^2-3}}{3x+2} \times \frac{1/|x|}{1/|x|}$

$= \lim_{x \to +\infty} \frac{\sqrt{x^2-3}}{3x+2} \times \frac{1/\sqrt{x^2}}{1/x}$; $|x| = \sqrt{x^2}$

$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

krn $x \to +\infty$, $|x| = x$

$= \lim_{x \to +\infty} \frac{\sqrt{\frac{x^2-3}{x^2}}}{\frac{3x+2}{x}}$

$= \lim_{x \to +\infty} \frac{\sqrt{1-\frac{3}{x^2}}}{3+\frac{2}{x}} = \frac{\lim_{x \to +\infty} \sqrt{1-\frac{3}{x^2}}}{\lim_{x \to +\infty} (3+\frac{2}{x})}$

cek lim peny.

$\lim_{x \to +\infty} (3+\frac{2}{x}) = 3 \neq 0$

$= \frac{\sqrt{\lim_{x \to +\infty} (1-\frac{3}{x^2})}}{3} = \frac{1}{3}$

② $\lim_{x \to +\infty} (\sqrt{x^6+3} - x^3) \times \frac{\sqrt{x^6+3}+x^3}{\sqrt{x^6+3}+x^3}$

$\lim_{x \to +\infty} \frac{x^6+3-x^6}{\sqrt{x^6+3}+x^3}$

$= \lim_{x \to +\infty} \frac{3}{\sqrt{x^6+3}+x^3} \times \frac{1/\sqrt{x^6}}{1/\sqrt{x^6}}$

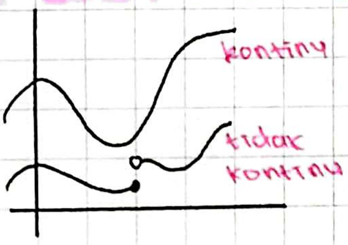
$\sqrt{x^6} = \sqrt{(x^3)^2} = |x^3|$

saat $x \to +\infty$, $\sqrt{x^6} = x^3$

$= \lim_{x \to +\infty} \frac{3/x^3}{\sqrt{1+3/x^6} + \frac{x^3}{x^3}}$

$= \frac{0}{\sqrt{1}+1} = 0$

KEKONTINUAN TURUNAN

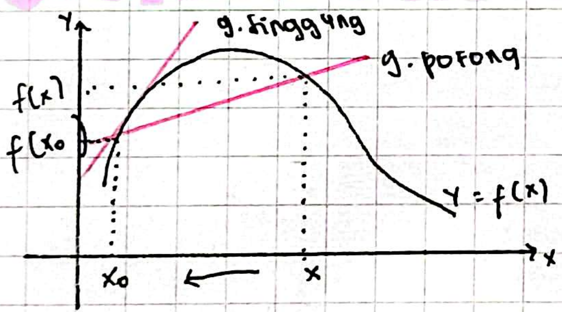


- fungsi f kontinu di $x = c$, jika:
  1. $f(c)$ terdefinisi
  2. $\lim\limits_{x \to c} f(x)$ ada
  3. $\lim\limits_{x \to c} = f(c)$

- fungsi f kontinu pada selang tertutup $[a,b]$, jika:
  1. f kontinu pada (a,b)
  2. f kontinu dari kanan di a
  3. f kontinu dari kiri di b

• Setiap polinomial kontinu dimana²
• Setiap fungsi rasional kontinu dimana² kec di titik² yg membuat peny. 0

G. Singgung



laju perubahan rata²
Kemiringan Garis Potong

$$m_{gp} = \frac{f(x) - f(x_0)}{x - x_0}$$

laju perubahan sesaat
Kemiringan Garis Singgung

$$m_{gs} = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

$$\boxed{f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}}$$

(hal. 115)

- $(uv)' = u'v + uv'$
- $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$

ex: ① $\frac{d}{dx}\left[(3x^2+7x-3)^{-6}\right]$

misal $u = 3x^2 + 7x - 3$

$$= \frac{d}{dx}(u^{-6}) = \frac{d}{du}(u^{-6}) \cdot \frac{du}{dx}$$

$$= -6(u^{-7}) \cdot \underbrace{(6x+7)}_{\text{turunan } u}$$

$$= -6(3x^2+7x-3)^{-7}(6x+7)$$

TURUNAN IMPLISIT

ex: ① $y = x^2 \to y' = \frac{dy}{dx} = 2x$

$y = f(x)$ $\quad \frac{d}{dx}[y]$

② $x^2 + y^2 + 1 = 0$

$$\frac{d}{dx}[x^2+y^2+1] = \frac{d}{dx}[0]$$

$$\frac{d}{dx}[x^2] + \frac{d}{dx}[y^2] + \frac{d}{dx}[1] = 0$$

$$2x + 2y\frac{dy}{dx} = 0$$

$$2y\frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = \frac{-2x}{2y} = -\frac{x}{y}$$

③ $xy^2 + x^2y = x$. Dapatkan $\frac{dy}{dx}$!

$$\frac{d}{dx}[xy^2 + x^2y] = \frac{d}{dx}[x]$$

$$\frac{d}{dx}[xy^2] + \frac{d}{dx}[x^2y] = 1$$

$u'v + uv'$

$$1 \cdot y^2 + 2y \cdot \frac{dy}{dx} \cdot x + 2xy + 1 \cdot \frac{dy}{dx} \cdot x^2 = 1$$

$$y^2 + 2xy\frac{dy}{dx} + 2xy + x^2\frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1 - y^2 - 2xy}{2xy + x^2}$$

④ $x^3 + y^3 = 3xy$. Dapatkan $\frac{dy}{dx}$!

$$\frac{d}{dx}[x^3+y^3] = \frac{d}{dx}[3xy]$$

$$3x^2 + 3y^2\frac{dy}{dx} = 3y + 1 \cdot \frac{dy}{dx} \cdot 3x$$

$$\frac{dy}{dx} = \frac{3y - 3x^2}{3y^2 - 3x}$$

(next dibalik dt aditenu)

15.) l. $2 \le \frac{x^2+1}{x} < x+3$

I. $2 \le \frac{x^2+1}{x}$

$= \frac{x^2+1}{x} \ge 2$

$\frac{x^2+1-2x}{x} \ge 0$

$\frac{(x-1)^2}{x} \ge 0$

$-$ | 0 | $+$ | 1 | $+$

Hp: $x > 0$

$(-\infty, 0)$

II. $\frac{x^2+1}{x} < x+3$

$\frac{x^2+1-(x^2+3x)}{x} < 0$

$\frac{1-3x}{x} < 0$

$-$ | 0 | $+$ | $1/3$ | $-$

Hp: $(-\infty,0) \cup [1/3, +\infty]$

Digabung

0 | 1 | $1/3$

Hp $= \{x \mid x > 1/3\}$

$= (1/3, +\infty)$

16) a. $\frac{5x-8}{2x-6} \le 2$

$= \frac{5x-8-4x+12}{2x-6} \le 0$

$= \frac{x+4}{2x-6} \le 0$

pembuat nol $x = -4 \vee x = 3$

$+$ | -4 | $-$ | 3 | $+$

Hp $= \{x \mid -4 \le x < 3\}$

$= [-4, 3)$

16.) b. $\frac{1}{x-5} \ge 3$

$= \frac{1-3x+15}{x-5} \ge 0$

$= \frac{16-3x}{x-5} \ge 0$

Pembuat 0; $x = 16/3$ & $x = 5$

$-$ | 5 | $+$ | $16/3 = 5,...$ | $-$

Hp $= \{x \mid 5 < x \le 16/3\}$

$= (5, 16/3]$

16) c. $\frac{3}{2-x} \ge \frac{4}{2x-3}$

$= \frac{3}{2-x} - \frac{4}{2x-3} \ge 0$

$= \frac{6x-9-8+4x}{(2-x)(2x-3)} \ge 0$

$= \frac{10x-17}{(2-x)(2x-3)} \ge 0$

pembuat nol $x = 17/10$

$x = 2$, $x = 3/2$

$+$ | $3/2$ (1,5) | $-$ | $17/10$ (1,7) | $+$ | 2 | $-$

$\frac{+}{(+)(-)}$ $\frac{-}{(+)(+)}$ $\frac{+}{(+)(+)}$ $\frac{+}{(-)(+)}$

Hp $= (-\infty, 3/2) \cup [17/10, 2)$

15.) j. $\frac{x+1}{2-x} \ge \frac{x}{x+3}$

$= \frac{x+1}{2-x} - \frac{x}{x+3} \ge 0$

$= \frac{x^2+4x+3-2x+x^2}{(2-x)(x+3)} \ge 0$

$= \frac{2x^2+2x+3}{(2-x)(x+3)} \ge 0$

g bisa di faktorkan maka tdk perlu dicari pembuat nol-nya. cukup penyebut s

$-$ | -3 | $+$ | 2 | $-$

$\frac{+}{(+)(-)}$ $\frac{+}{(+)(+)}$ $\frac{+}{(-)(+)}$

Hp $= (-3, 2)$

⑤ $y^2 - x + 1 = 0$. kemiringan garis singgung di titik $(2,-1)$ & $(2,1)$

$$\frac{d}{dx}\left[y^2 - x + 1\right] = \frac{d}{dx}[0]$$

$$2y \cdot \frac{dy}{dx} - 1 = 0$$

$$\frac{dy}{dx} = \frac{1}{2y}$$

$$\left.\frac{dy}{dx}\right|_{x=2,\ y=-1} = \frac{1}{2(-1)} = -\frac{1}{2}$$

$$\left.\frac{dy}{dx}\right|_{x=2,\ y=1} = \frac{1}{2 \cdot 1} = \frac{1}{2}$$

(lanjutan soal no. 4)

④ b) pers. garis singgung di $\left(\frac{3}{2}, \frac{3}{2}\right)$.

$$\left.\frac{dy}{dx}\right|_{x=3/2,\ y=3/2} = \frac{3\left(\frac{3}{2}\right) - 3\left(\frac{3}{2}\right)^2}{3\left(\frac{3}{2}\right)^2 - 3\left(\frac{3}{2}\right)} = \frac{\frac{9}{4} - \frac{27}{2}}{\frac{27}{2} - \frac{9}{4}} = \frac{\frac{9}{4} - \frac{27}{2}}{-1\left(\frac{9}{4} - \frac{27}{2}\right)} = -1$$

PGS. $\left(\frac{3}{2}, \frac{3}{2}\right)$; kemiringan $= -1$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{3}{2} = -1\left(x - \frac{3}{2}\right)$$

$$y = -x + \frac{3}{2} + \frac{3}{2}$$

$$y = -x + 3$$

⑥ Dapatkan titik singgung pada kurva dgn garis singgung horisontal → $m = 0$ → $\frac{dy}{dx} = 0$

$$\frac{3y - 3x^2}{3y^2 - 3x} = 0$$

$$3y - 3x^2 = 0$$

$$y - x^2 = 0$$

$$y = x^2$$

masukkan ke pers. awal

$$x^3 + (x^2)^3 = 3x(x^2)$$

$$x^3 + x^6 = 3x^3$$

$$x^3 - 3x^3 + x^6 = 0$$

$$x^6 - 2x^3 = 0$$

$$x^3(x^3 - 2) = 0$$

① $x^3 = 0$, $x = 0$, $y = x^2 = 0$

② $x^3 - 2 = 0$, $x = \sqrt[3]{2} = 2^{\frac{1}{3}}$, $y = x^2 = 2^{\frac{2}{3}}$

ada 2 titik

⑥ $\frac{d}{dx}\left[\sqrt{x^3 + \csc x}\right]$

$$= \frac{d}{dx}\left[(x^3 + \csc x)^{\frac{1}{2}}\right]$$

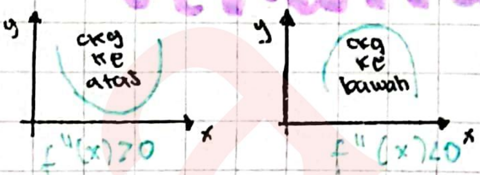
$$= \frac{1}{2}(x^3 + \csc x)^{-\frac{1}{2}}(3x^2 - \csc \cdot \cot x)$$

$$= \frac{3x^2 - \csc \cdot \cot x}{2\sqrt{x^3 + \csc x}}$$

FUNGSI NAIK TURUN

- naik → $f'(x) > 0$
- turun → $f'(x) < 0$
- konstan → $f'(x) = 0$ titik stasioner

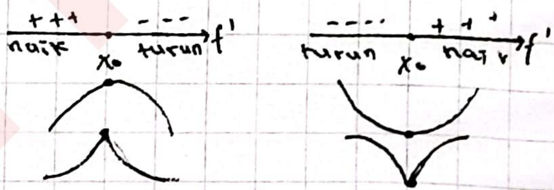
KE-CEKUNGan



titik belok = $f''(x) = 0$

EKSTRIM RELATIF

Uji turunan pertama



Uji turunan kedua

maks relatif | min relatif

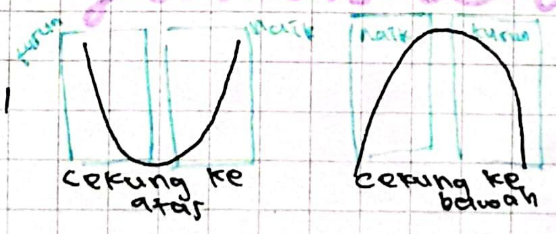
di x_0

$f''(x_0) < 0$ (ckg bawah) | substitusi t. kritis ke f'' | $f''(x_0) > 0$ (ckg atas)

ER trjd pd titik kritis

1. titik stasioner : $f'(x) = 0$
1. titik dmn turunan pertama tak ada.

gambar



GRAFIK POLINOMIAL

ex: ① $y = x^3 - 3x + 2$

↳ $y' = 3x^2 - 3$

$0 = 3x^2 - 3$

$0 = x^2 - 1$

$0 = (x+1)(x-1)$

+ + + naik (−1) − − − turun (1) + + + naik → $(-1, 4)$, $(1, 0)$

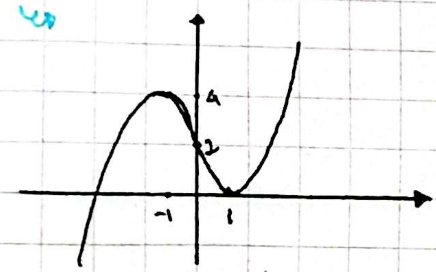
↳ $y'' = 6x$

$0 = 6x$

$x = 0$

− − − cekung ke bawah (0) + + + cekung ke atas → $(0, 2)$

↳



GRAFIK FUNGSI RASIONAL

ex: ① $y = \dfrac{2x^2 - 8}{x^2 - 16}$

I. Simetri trhdp sb. Y

x diganti $-x$ ✓ (krn x^2)

II. Perpotongan trhdp sb. x

$Y = 0$ → $(-2, 0)$ dan $(2, 0)$

$0 = \dfrac{2x^2 - 8}{x^2 - 16}$

$0 = 2x^2 - 8$

$0 = x^2 - 4$

$0 = (x+2)(x-2)$

$x = -2 \vee x = 2$

III. Perpotongan trhdp sb. Y

$x = 0$

$Y = -8/-16 = \frac{1}{2}$ → $(0, \frac{1}{2})$

IV. Asimtot tegak — tidak akan memotong AT krn penyebut = 0

$x^2 - 16 = 0$

$x^2 = 16$

$x = \pm 4$

V. Asimtot datar — bisa memotong AD.

ingin tahu nilai y saat x sangat kecil ∨ besar

↳ $\lim_{x \to +\infty} \dfrac{2x^2 - 8}{x^2 - 16} = \dfrac{2x^2}{x^2} = 2$

↳ $\lim_{x \to -\infty} \dfrac{2x^2 - 8}{x^2 - 16} = \dfrac{2x^2}{x^2} = 2$

Jadi AD $y = 2$

VI. Turunan

↳ $y' = \dfrac{u'v - uv'}{v^2}$

$= \dfrac{-48x}{(x^2 - 16)^2}$ ⊕

↳ $y'' = \dfrac{-48(x^2-16)^2 - ((-48x) \cdot 2 \cdot (x^2-16) \cdot 2x)}{(x^2-16)^{4\,3}}$

$= \dfrac{-48(x^2-16) - [(-48x) \cdot 2 \cdot 2x]}{(x^2-16)^3}$

$= \dfrac{-48(x^2 - 16 - 4x^2)}{(x^2-16)^3}$

$= \dfrac{48(3x^2 + 16)}{(x^2-16)^3}$ ⊕

→ y'

• pembilang nol: $-48x = 0$, $x = 0$

• penyebut nol: $(x^2-16)^2 = 0$, $x^2 - 16 = 0$ → $x = \pm 4$

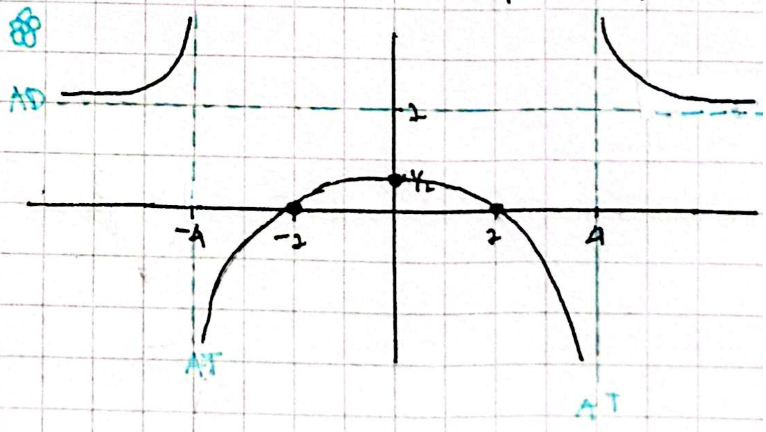
→ y''

• pembilang nol: Tak Mungkin

• penyebut nol: $(x^2-16)^3 = 0$, $x^2 - 16 = 0$ → $x = \pm 4$

y': + + naik (−4) + + (0) − − turun (4) − −

y'': + + c. atas (−4) − − c. bawah (4) + + c. atas



① →

① $y = \frac{x^2-2}{x}$

I. Simetri trhdp sb. Y

tidak.

II. Perpotongan sb. x

$0 = \frac{x^2-2}{x}$

$0 = x^2-2$

$x = \pm\sqrt{2} \rightarrow (-\sqrt{2},0)\ (\sqrt{2},0)$

III. Perpotongan sb. Y

$y = \frac{-2}{0}$?

IV. Asimtot tegak

$x = 0$

V. Asimtot miring — derajat pembilang > derajat penyebut (2>1)

$\frac{x^2-2}{x} = x + \frac{-2}{x}$

AM: $y = x$

VI. Turunan

$y' = \frac{2x(x)-(x^2-2)\cdot 1}{x^2}$

$= \frac{2x^2-x^2+2}{x^2} = \frac{x^2+2}{x^2}$

$y'' = \frac{2x(x^2)-(x^2+2)(2x)}{x^4}$

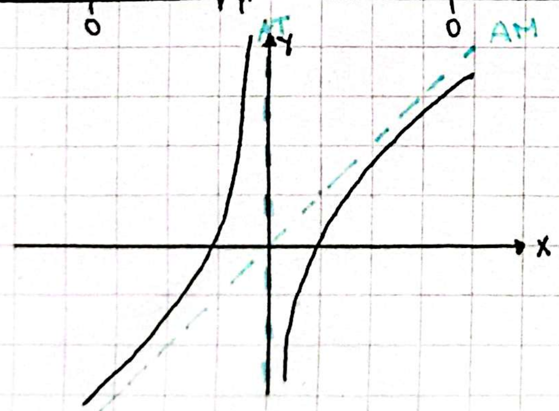
$= \frac{2x^3-2x^3-4x}{x^4} = \frac{-4}{x^3}$

→ y' • pembilang nol: $x^2+2=0$, TM • penyebut nol: $x^2=0$, $x=0$

→ y'' • pembilang nol: TM • penyebut nol: $x^3=0$, $x=0$

y': ++ naik ++ (0)

y'': ++ c. atas, −− c. bawah (0)



EKSTRIM ABSOLUT

ex: ① Tentukan nilai maks & min $f(x) = 2x^3 - 15x^2 + 36x$ pd interval $[1,5]$

$f'(x) = 0$

$6x^2 - 30x + 36 = 0$

$x^2 - 5x + 6 = 0$

$(x-2)(x-3) = 0$

$x = 2 \lor x = 3$

x	1	2	3	5
f(x)	23 (min)	35	27	55 (max)

② Tentukan nilai ekstrim

$f(x) = 6x^{4/3} - 3x^{1/3}$ pd interval $[-1,1]$

$f'(x) = 6\cdot\frac{4}{3}\cdot x^{1/3} - 3\cdot\frac{1}{3}\cdot x^{-2/3}$

$= 8x^{1/3} - \frac{1}{x^{2/3}}$

$= \frac{8x-1}{x^{2/3}}$

titik stasioner

$f'(x) = 0$

$\frac{8x-1}{x^{2/3}} = 0$

$8x-1 = 0$

$x = 1/8$

turunan pertama tidak ada → peny = 0

$x^{2/3} = 0$

$x = 0$

x	−1	0	1/8	1
f(x)	9	0	−9/8	3

③ INTERVAL TAK HINGGA

Tentukan apakah fungsi $f(x) = 3x^4 + 4x^3$ mempunyai nilai maks or min. Jika punya, cari nilainya.

$\lim_{x\to-\infty} 3x^4+4x^3 = \lim_{x\to-\infty} 3x^4 = +\infty$

$\lim_{x\to+\infty} 3x^4+4x^3 = \lim_{x\to+\infty} 3x^4 = +\infty$

f hanya punya min

$f'(x) = 0$

$12x^3 + 12x^2 = 0$

$12x^2(x+1) = 0$

$x = 0 \lor x = -1$

−− (−1, min) ++ (0) ++

$f(-1) = 3 + (-4) = -1$

∴ nilai min f adalah −1 yg terjadi saat $x = -1$

$\lim_{x\to-\infty} f(x)$	$\lim_{x\to+\infty} f(x)$	Kesimpulan
$+\infty$	$+\infty$	min
$-\infty$	$-\infty$	maks
$-\infty$	$+\infty$	—
$+\infty$	$-\infty$	—

① Tentukan nilai maks & min dari $f(x) = \frac{1}{x^2-x}$ pd interval tebuka

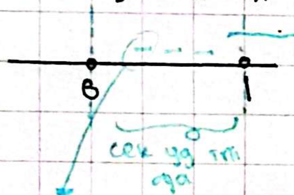
$(0,1)$

$x=0$, $x=1$ → penyebut 0

$\lim_{x\to0^+} \frac{1}{x^2-x} = -\infty$

$\lim_{x\to1^-} \frac{1}{x^2-x} = -\infty$

maka hanya punya nilai maks



cara

$f(\frac{1}{2}) = \frac{1}{\frac{1}{4}-\frac{1}{2}} = \frac{1}{(-)} = (-)$

$f(x) = (x^2-x)^{-1}$

$f'(x) = -1(x^2-x)^{-2}\cdot(2x-1)$

$= \frac{1-2x}{(x^2-x)^2}$

Titik stasioner

$f'(x)=0$

$1-2x=0$

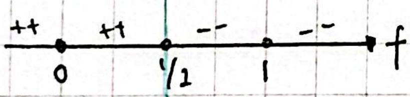
$x = \frac{1}{2}$

turunan pertm pada

$(x^2-x)^2=0$

$x(x-1)=0$

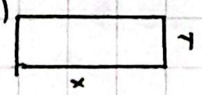
$x=0 \quad \vee \quad x=1$



nilai maks $= f(\frac{1}{2}) = \frac{1}{\frac{1}{4}-\frac{1}{2}} = \frac{1}{-\frac{1}{4}} = -4$

APLIKASI TURUNAN

1.)



$K = 100$

x & y spy L maks ?

① $2(x+y) = 100$

$x+y = 50$

$y = 50-x$

② $L = x\cdot y$

$= x(50-x)$

$L(x) = 50x - x^2$

③ $x \geq 0$

$y \geq 0$

$50 - x \geq 0$

$50 \geq x \to x \leq 50$

maka $0 \leq x \leq 50$

$[0,50]$

④ $L'(x) = 0$

$50-2x = 0$

$2x = 50$

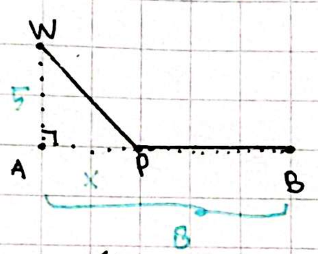
$x = 25$

⑤

x	0	25	50
L	0	625	0

max

2.



biaya minimum = ?

$PW = \sqrt{25+x^2}$

$PB = 8-x$

$0 \leq x \leq 8$

$C(x) = (\sqrt{25+x^2})\cdot 100 + (8-x)\cdot 75$

$C'(x) = 100\cdot\frac{1}{2}(25+x^2)^{-1/2}\cdot 2x + (-75)$

$= \frac{100x}{\sqrt{25+x^2}} - 75 = 0$

$\frac{4\,100x}{\sqrt{25+x^2}} = 75 \;3$

$16x^2 = 9(25+x^2)$

$16x^2 = 225 + 9x^2$

$7x^2 = 225$

$x = +\sqrt{\frac{225}{7}} \;\checkmark \qquad -\sqrt{\frac{225}{7}}$ TM

x	0	$15/\sqrt{7}$	8
C	1100	930,...	943,...

min.

Saat P $15/\sqrt{7} \approx 5{,}67$ mn dari A

integral

I. Integral tak tentu :

$\int x \, dx = \frac{1}{2} x^2 \boxed{+C}$

$$\boxed{\int x^n \, dx = \frac{x^{n+1}}{n+1} + C}$$

II. Integral tertentu :

$\int_1^2 x \, dx = \text{angka}$

trigonometri

1. $\frac{d}{dx} \sin x = \cos x \rightarrow \int \cos x \, dx = \sin x + c$

2. $\frac{d}{dx} \cos x = -\sin x \rightarrow \int \sin x \, dx = -\cos x + c$

3. $\frac{d}{dx} \tan x = \sec^2 x \rightarrow \int \sec^2 x \, dx = \tan x + c$

4. $\frac{d}{dx} \cot x = -\csc^2 x \rightarrow \int \csc^2 x \, dx = -\cot x + c$

5. $\frac{d}{dx} \sec x = \sec x \cdot \tan x \rightarrow \int \sec x \cdot \tan x \, dx = \sec x + c$

6. $\frac{d}{dx} \csc x = -\csc \cdot \cot x \rightarrow \int -\csc \cdot \cot x \, dx = \csc x + c$

Dst....

notes :

- $1/\sin x = \csc x$
- $1/\cos x = \sec x$
- $\sin x / \cos x = \tan x$
- $\cos x / \sin x = \cot x$
- $\sin^2 x + \cos^2 x = 1$
- $\tan^2 x + 1 = \sec^2 x$
- $\sin 2x = 2 \sin \cdot \cos$
- $\cos 2x = \cos^2 x - \sin^2 x$
  $= 2\cos^2 x - 1$
  $= 1 - 2\sin^2 x$

Latso Hal. 218

12. $\int \left(\frac{7}{y^{3/4}} - \sqrt[3]{y} + 4\sqrt{y} \right) dy$

$= \int \left(7 y^{-3/4} - y^{1/3} + 4 y^{\frac{1}{2}} \right) dy$

$= \frac{7 y^{1/4}}{1/4} - \frac{y^{4/3}}{4/3} + \frac{4 y^{3/2}}{3/2} + C$

$= 28 y^{1/4} - \frac{3}{4} y^{4/3} + \frac{8}{3} y^{3/2} + C$

18. $\int \frac{\sin x}{\cos^2 x} \, dx$

$= \int \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} \, dx$

$= \int \tan x \cdot \sec x \, dx$

$= \sec x + c$

30. $\int \frac{1}{1 + \sin x} \, dx \cdot \frac{1 - \sin x}{1 - \sin x}$

$= \int \frac{1 - \sin x}{1 - \sin^2 x} \, dx$

$= \int \frac{1 - \sin x}{\cos^2 x} \, dx$

$= \int \frac{1}{\cos^2 x} - \frac{\sin x}{\cos^2 x} \, dx$

$= \int \frac{1}{\cos^2 x} \, dx - \int \frac{\sin x}{\cos^2 x} \, dx$

$= \int \sec^2 x \, dx - \int \tan x \cdot \sec x \, dx$

$= \tan x - \sec x + c$

dengan SUBTITUSI

1. $\int (x-8)^{23} \, dx$

misal $u = x - 8$

$\frac{du}{dx} = 1 \rightarrow dx = du$

$= \int u^{23} \cdot du = \frac{u^{24}}{24} + C$

$= \frac{(x-8)^{24}}{24} + C$

2. $\int \sin(2x + 9) \, dx$

misal $u = 2x + 9$

$du = 2dx \rightarrow dx = \frac{1}{2} du$

$= \frac{1}{2} \int \sin u \cdot du$

$= -\frac{1}{2} \cos u + c$

$= -\frac{1}{2} \cos (2x + 9) + c$

3. $\int \sin^2 x \cdot \cos x \, dx$

misal $u = \sin x$

$du = \cos x \, dx$

$= \int u^2 \cdot du$

$= \frac{1}{3} \cdot u^3 + C$

$= \frac{\sin^3 x}{3} + C$

4. $\int \frac{\cos \sqrt{x}}{\sqrt{x}} \, dx$

misal $\cdot\ u = \sqrt{x} = x^{1/2}$

$\cdot\ du = \frac{1}{2} x^{-1/2} \, dx$

$du = \frac{1}{2\sqrt{x}} \, dx \rightarrow \frac{dx}{\sqrt{x}} = 2\,du$

$= 2 \int \cos u \cdot du$

$= 2 \sin u + C$

$= 2 \sin \sqrt{x} + C$

5. $\int x^2 \sqrt{x-1} \, dx$

misal $u = x - 1 \rightarrow \boxed{x = u + 1}$

$du = dx$

$= \int (u+1)^2 \, u^{1/2} \, du$

$= \int (u^2 + 2u + 1) \cdot u^{1/2} \, du$

$= \int u^{5/2} + 2u^{3/2} + u^{1/2} \, du$

$= \frac{u^{7/2}}{7/2} + \frac{2u^{5/2}}{5/2} + \frac{u^{3/2}}{3/2} + C$

$= \frac{7}{2} u^{7/2} + \frac{4}{5} u^{5/2} + \frac{2}{3} u^{3/2} + C$

$= \frac{7}{2} (x-1)^{7/2} + \frac{4}{5} (x-1)^{5/2} + \frac{2}{3} (x-1)^{3/2} + C$

6. $\int \frac{\sin 2\theta}{(5 + \cos 2\theta)^{3/2}} \, d\theta$

misal $\cdot\ u = 5 + \cos 2\theta$

$\cdot\ du = 0 + 2(-\sin 2\theta) d\theta$

$\hookrightarrow \sin 2\theta \cdot d\theta = -\frac{1}{2} du$

$= -\frac{1}{2} \int u^{-3/2} \, du$

$= \frac{1}{-2} \cdot \frac{u^{-1/2}}{-1/2} + C$

$= \frac{1}{\sqrt{u}} + C$

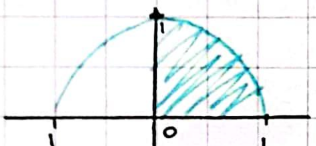
$= \frac{1}{\sqrt{5 + \cos 2\theta}} + C$

INTEGRAL dengan rumus LUAS

ex: ① $\int_0^1 \sqrt{1-x^2}\,dx$

→ $y^2 = 1-x^2$

$y^2 + x^2 = 1$ (pers. lingk dgn pusat (0,0) namun y (+))



→ $L = \frac{1}{4}\pi r^2$

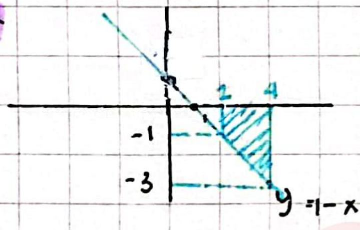
$= \frac{1}{4}\pi \cdot 1^2$

$= \pi/4$

② $\int_2^4 (1-x)\,dx$

→ $y = 1-x$

x	0	1
y	1	0



→ $L = \frac{(1+3)\cdot 2}{2}$

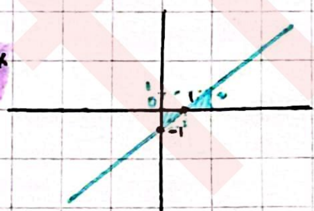
$= 4$

di (-) kan krn daerah berada di bawah sb.x

∴ $L = -4$

③ $\int_0^2 (x-1)\,dx$

x	0	1
y	-1	0



→ L atas − L bawah

krn L-nya sama, maka $L = 0$

NOTES!

$\int_1^0 \sqrt{1-x^2}\,dx = -\int_0^1 \sqrt{1-x^2}\,dx$

$\int_a^b c\cdot f(x)\,dx = c\cdot\int_a^b f(x)\,dx$

$\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$

$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

TEOREMA FUNDAMENTAL

ex: ① $\int_1^2 x\,dx =$

$= \left.\frac{x^2}{2}\right]_1^2$

$= \frac{1}{2}(2^2 - 1^2)$

$= 3/2$

② hal. 25:

$f(x) = \begin{cases} x^2, & x < 2, \\ 3x-2, & x \geq 2. \end{cases}$

$\int_0^6 f(x) = ?$

→ $\int_0^6 f(x)\,dx =$

$= \int_0^2 x^2\,dx + \int_2^6 (3x+2)\,dx$

$= \left.\frac{x^3}{3}\right]_0^2 + \left.\frac{3x^2}{2} - 2x\right]_2^6$

$= \frac{1}{3}(2^3 - 0^3) + \frac{3}{2}(6^2 - 2^2) - 2(6-2)$

$= \frac{8}{3} + 48 - 8 = \frac{8}{3} + 40 = \frac{128}{3}$

③ $\int_{-1}^2 |x|\,dx = ?$

→ $|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$

→ $\int_{-1}^2 |x|\,dx =$

$= \int_{-1}^0 -x\,dx + \int_0^2 x\,dx$

$= \left.\frac{-x^2}{2}\right]_{-1}^0 + \left.\frac{x^2}{2}\right]_0^2$

$= -\frac{1}{2}(0^2 - (-1)^2) + \frac{1}{2}(2^2 - 0^2)$

$= \frac{1}{2} + 2 = 5/2$

F rata

$f_{rata} = \frac{1}{b-a}\int_a^b f(x)\,dx.$

ex: ① f rata pada interval [1, 4] dari $f(x) = x^2$?

→ $f(x^*) = \frac{1}{4-1}\int_1^4 x^2\,dx$

$= \frac{1}{3}\cdot\left.\frac{x^3}{3}\right]_1^4$

$= \frac{1}{3}\cdot\frac{1}{3}(4^3 - 1^3)$

$= \frac{1}{9}\cdot 63 = 7$

ex: ① $\int_0^2 x(x^2+1)^3\,dx. = ?$

↳ • $u = x^2+1$

• $du = 2x\,dx$

$\frac{1}{2}\,du = x\,dx$

↳ $x=0 \to u = 0^2+1 = 1$

$x=2 \to u = 2^2+1 = 5$

↳ $\int_1^5 u^3 \, \frac{1}{2}\,du$

$= \frac{1}{2} \int_1^5 u^3\,du$

$= \frac{1}{2} \cdot \frac{u^4}{4}\Big]_1^5$

$= \frac{1}{2}\cdot\frac{1}{4}\left(5^4-1^4\right)$

$= \frac{1}{8} \cdot 624 = 78$

② $\int_0^{\pi/4} \cos(\pi - x)\,dx = ?$

↳ $u = \pi - x$

$du = -1\,dx$

↳ $x=0 \to u = \pi - 0 = \pi$

$x = \pi/4 \to u = \pi - \pi/4 = 3\pi/4$

↳ $\int_\pi^{3\pi/4} -\cos u\,du$

$= -\int_\pi^{3\pi/4} \cos u\,du$

$= -\left(\sin u\Big]_\pi^{3\pi/4}\right)$

$= -\left(\sin 3\pi/4 - \sin \pi\right)$

$= -\left(\frac{1}{2}\sqrt{2} - 0\right) = -\frac{1}{2}\sqrt{2}$

③ $\int_0^{\pi/8} \sin^5 2x \cdot \cos 2x \cdot dx = ?$

↳ • $u = \sin 2x$

• $du = -2\cos 2x\,dx$

$-\frac{1}{2}\,du = \cos 2x\,dx$

↳ $x=0 \to u = \sin 2\cdot 0 =$

$x = \pi/6 \to u = \sin 2\cdot\pi/6 =$

Eigen

ex :

$A = \begin{pmatrix} 2 & 0 \\ 3 & -1 \end{pmatrix} \qquad x = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$Ax = \begin{pmatrix} 2 & 0 \\ 3 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 2 \\ 2 \end{pmatrix}}_{\text{kelipatan}}$

$Ax = \lambda x$

$Ax = \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \lambda x$

nilai eigen (2)

vektor eigen $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

NILAI EIGEN

1. $|A - \lambda I| = 0$

$\left| \begin{pmatrix} 2 & 0 \\ 3 & -1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$

$\left| \begin{pmatrix} 2 & 0 \\ 3 & -1 \end{pmatrix} + \begin{pmatrix} -\lambda & 0 \\ 0 & -\lambda \end{pmatrix} \right| = 0$

$\begin{vmatrix} 2-\lambda & 0 \\ 3 & -1-\lambda \end{vmatrix} = 0$

$(2-\lambda)(-1-\lambda) - (3 \cdot 0) = 0$

$(2-\lambda)(-1-\lambda) = 0$

$2-\lambda = 0 \qquad -1-\lambda = 0$

$\lambda = 2 \qquad \lambda = -1$

↳ Vektor eigen saat $\lambda = 2$

$(A - \lambda I)x = 0$

$\begin{bmatrix} 0 & 0 \\ 3 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$3x_1 - 3x_2 = 0$

$3x_1 = 3x_2$

$x_1 = x_2 = t$

→ $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} t \\ t \end{pmatrix} = t \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ vektor eigen

↳ Vektor eigen saat $\lambda = -1$

$(A - \lambda I)x = 0$

$\begin{bmatrix} 3 & 0 \\ 3 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$3x_1 = 0 \rightarrow x_1 = 0$ → $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ t \end{pmatrix} = t \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$x_2 = t.$

* cek :

$A \begin{pmatrix} 2 & 0 \\ 3 & -1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix}_{\text{vek. eigen}} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}^{\lambda \cdot x}$

kelipatan ✓ $Ax = \lambda x$

2. $A = \begin{pmatrix} 3 & 0 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$

$|A - \lambda I| = 0$

$\begin{vmatrix} 3-\lambda & 0 & 1 \\ 1 & 2-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$

pake sarrus bs, inipake kofaktor baris 3

$(2-\lambda) \cdot [(3-x)(2-\lambda) - 0 \cdot 1] = 0$

$(2-\lambda)(3-\lambda)(2-\lambda) = 0$

$\lambda = 2 \qquad \lambda = 3$

↳ Vektor eigen saat $\lambda = 2$

$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] b_2 - b_1 \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$x_1 + x_3 = 0$

$x_1 = -x_3 = -t$

$x_3 = t$

$x_2 = s$

→ $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -t \\ s \\ t \end{pmatrix} = t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + s \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ V.E.

↳ Vektor eigen saat $\lambda = 3$

$\begin{bmatrix} 0 & 0 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right]$ tukar baris $b_1 \leftrightarrow b_2$

$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right] \begin{array}{l} b_2 - b_1 \\ \\ b_3 + b_1 \end{array}$

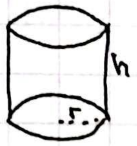
$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ → $x_3 = 0$, $x_1 = x_2 = t$

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} t \\ t \\ 0 \end{pmatrix} = t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

asistensi

Hal. 142

3.)



h & r fungsi waktu

a. Hub. $\frac{dV}{dt}$, $\frac{dh}{dt}$, & $\frac{dr}{dt}$?

↳ $V = \pi r^2 h \quad \to u' = \frac{d}{dt}\pi r^2 = 2\pi r\frac{dr}{dt}$

$v' = \frac{d}{dt} h = 1 \cdot \frac{dh}{dt}$

$V' = u'v + uv'$

$\frac{dV}{dt} = 2\pi r h \frac{dr}{dt} + \pi r^2 \frac{dh}{dt}$

b. $h = 6\,cm$ $\quad \frac{dh}{dt} = 1\,cm/det$

$r = 10\,cm$ $\quad \frac{dr}{dt} = -1\,cm/det$

↳ $\frac{dV}{dt} = 2\pi \cdot 6 \cdot 10 \cdot (-1) + \pi \cdot (10)^2 \cdot 1$

$= \pi(-120 + 100)$

$= -20\pi\ cm^3/det$

2.) r fungsi dari waktu. A = Luas ⊙

a. Hub. $\frac{dA}{dt}$ & $\frac{dr}{dt}$?

↳ $A = \pi r^2$

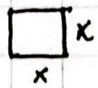
$A' = 2\pi r \frac{dr}{dt}$

$\frac{dA}{dt} = 2\pi r \cdot \frac{dr}{dt}$

b. $r = 5\,cm$. $\frac{dr}{dt} = 2\,cm/det$

$\frac{dA}{dt} = 2 \cdot \pi \cdot 5 \cdot 2 = 20\pi\ cm^2/det$

1.)



x fungsi dari waktu

L = Luas □

a. Hub $\frac{dL}{dt}$ & $\frac{dx}{dt}$?

↳ $L = x^2$

$\frac{dL}{dt} = 2x \cdot \frac{dx}{dt}$

b. $x = 3m$. $\frac{dx}{dt} = 2\,m/menit$

↳ $\frac{dL}{dt} = 2 \cdot 3 \cdot 2 = 12\ m^2/menit$

Hal. 154

5.) $f(x) = x^2 - 3x + 8$

↳ $f'(x) = 2x - 3 = 0$

$2x = 3$

$x = 3/2$

turun: − − − | naik: + + + (di 1,5)

a. $(1{,}5, +\infty)$

b. $(-\infty, 1{,}5)$

↳ $f''(x) = 2 > 0$

c. tidak ada

d. cekung ke atas pada $(-\infty, +\infty)$

e. tidak ada titik belok

7.) $f(x) = (2x+1)^3$

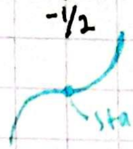
↳ $f'(x) = 3(2x+1)^2 \cdot 2$

$= 6(2x+1)^2 = 0$

$(2x+1)^2 = 0$

$x = -1/2$

+ + | + + (di $-1/2$)



a. naik: $(-\infty, +\infty)$

b. turun: tidak ada

↳ $f''(x) = 12(2x+1) \cdot 2$

$24(2x+1) = 0$

$x = -1/2$

− − | + + (di $-1/2$)

c. c. bawah : $(-\infty, -1/2)$

d. c. atas : $(-1/2, +\infty)$

e. titik belok : $-1/2$